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API Automation Complete Guide

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10. Data-Driven API Testing
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API FUNDAMENTALS



API AUTOMATION → API FUNDAMENTALS



Topics Covered

1. What is API
 2. Types of APIs
 3. API Architecture
 4. Client-Server Communication
 5. REST API Basics ★
 6. JSON Basics
 7. Real-Time Flow
 8. Best Practices
-



1 What is API

1 What is it

API (Application Programming Interface) is a medium that allows two applications or systems to communicate with each other.

2 Key Components

- Request
 - Response
 - Endpoint
 - Client
 - Server
-

3 How YOU should answer (Interview Style)

API acts as a bridge between client and server. It allows applications to exchange data using requests and responses without directly accessing internal implementation.

4 Practical Example

Mobile App → API → Database

Example:

PhonePe app sends payment request to backend API.

5 Common Mistakes

- ✗ Saying API is only for testing
 - ✗ Confusing API with database
 - ✗ Ignoring request-response flow
-

6 Expected Interview Questions

- What is API?
 - Why are APIs used?
 - Real-time API example?
-

2 Types of APIs

1 What is it

Different API categories based on accessibility and architecture.

2 Key Components

- REST API
 - SOAP API
 - GraphQL API
 - Internal API
 - Public API
-

3 How YOU should answer (Interview Style)

REST APIs are most commonly used in modern applications because they are lightweight and use JSON for communication. SOAP APIs are XML-based and more strict.

4 Practical Example

REST API → JSON

SOAP API → XML

5 Common Mistakes

- ✗ Not knowing REST vs SOAP
 - ✗ Saying SOAP is modern standard
 - ✗ Ignoring GraphQL basics
-

6 Expected Interview Questions

- Difference between REST and SOAP?
 - What APIs have you worked on?
 - Why is REST popular?
-

3 API Architecture

1 What is it

API architecture defines how APIs are designed and communicate.

2 Key Components

- Client
 - Server
 - Endpoint
 - HTTP Protocol
 - Request/Response
-

3 How YOU should answer (Interview Style)

API architecture follows a client-server model where the client sends requests and the server processes and returns responses over HTTP protocol.

4 Practical Example

Client → HTTP Request → Server
Server → HTTP Response → Client

5 Common Mistakes

- ✗ Confusing frontend and backend roles
 - ✗ Not understanding stateless communication
-

6 Expected Interview Questions

- Explain API architecture.
 - What is client-server communication?
 - What protocol APIs use?
-

4 Client-Server Communication

1 What is it

Communication model where client sends requests and server returns responses.

2 Key Components

- Request
 - Response
 - HTTP
 - Status Code
 - Data Exchange
-

3 How YOU should answer (Interview Style)

In client-server communication, the client sends HTTP requests to the server, and the server processes the request and returns a response with status code and data.

4 Practical Example

Browser → Login Request → Server

Server → Authentication Response → Browser

5 Common Mistakes

- ✗ Ignoring status codes
 - ✗ Confusing request and response
-

6 Expected Interview Questions

- Explain request-response cycle.
 - What happens when API is called?
 - What is stateless communication?
-

5 REST API Basics

1 What is it

REST (Representational State Transfer) is a lightweight architecture used for API communication over HTTP.

2 Key Components

- HTTP methods
 - Stateless
 - JSON
 - Resource-based URLs
-

3 How YOU should answer (Interview Style)

REST APIs use HTTP methods like GET, POST, PUT, and DELETE for communication. They are lightweight, stateless, and mostly use JSON data format.

4 Practical Example

GET /users
POST /users
PUT /users/1
DELETE /users/1

5 Common Mistakes

- ✗ Not understanding statelessness
 - ✗ Confusing REST with HTTP
 - ✗ Ignoring resource-based design
-

6 Expected Interview Questions

- What is REST API?
 - Why is REST widely used?
 - What is a stateless API?
-

6 JSON Basics

1 What is it

JSON (JavaScript Object Notation) is a lightweight data format used in APIs.

2 Key Components

- Key-value pairs
 - Objects
 - Arrays
 - Nested JSON
-

3 How YOU should answer (Interview Style)

JSON is a lightweight and readable data format used for request and response communication in REST APIs.

4 Practical Example

```
{  
  "name": "Kalpesh",  
  "role": "SDET"  
}
```

5 Common Mistakes

- ✗ Invalid JSON syntax
 - ✗ Confusing object and array
-

6 Expected Interview Questions

- What is JSON?
 - Why is JSON preferred?
 - Difference between JSON object and array?
-

7 Real-Time API Flow

1 What is it

Understanding how APIs work in real applications.

2 Key Components

- Frontend
 - Backend
 - Database
 - API layer
-

3 How YOU should answer (Interview Style)

In real-time applications, frontend sends requests to backend APIs, APIs process business logic, interact with database, and return responses to frontend.

4 Practical Example

Mobile App
↓
API Layer
↓
Database

5 Common Mistakes

- ✗ Thinking frontend directly accesses database
 - ✗ Ignoring backend processing
-

6 Expected Interview Questions

- Explain real-time API flow.
 - How frontend communicates with backend?
 - Where APIs fit in architecture?
-

8 Best Practices

1 What is it

Guidelines for understanding and testing APIs effectively.

2 Key Components

- Proper validations
 - Understanding architecture
 - Response verification
 - Error handling
-

3 How YOU should answer (Interview Style)

I focus on understanding request-response flow, validating responses correctly, and using proper API structures for maintainable testing.

4 Practical Example

Validate:

- Status Code
 - Response Body
 - Headers
 - Response Time
-

5 Common Mistakes

- ✗ Only validating status codes
 - ✗ Ignoring negative testing
 - ✗ Weak validations
-

6 Expected Interview Questions

- API testing best practices?
 - What validations do you perform?
 - How do you test APIs effectively?
-

FINAL INTERVIEW GOLD

MOST IMPORTANT TOPICS

1. REST API ★
2. Request-Response Flow ★
3. JSON Basics ★
4. Real-Time Architecture ★